Leverage Causal Inference to Quantify the Survival Benefit of Multidisciplinary Care for Amyotrophic Lateral Sclerosis Using Real-World Data

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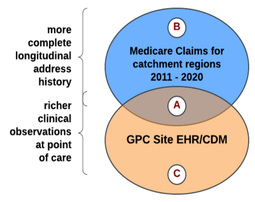
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Introduction

- Timely access to **multidisciplinary care (MDT)** is a critical component of the ALS care pathway, integrating neurology, rehabilitation, speech-language, respiratory, nutritional, and supportive care.
- Randomized trials of multidisciplinary versus less comprehensive care are ethically and practically challenging, as they could restrict access to clinically indicated services.
- Real-world evaluation is complicated by the dynamic, heterogeneous nature of MDT, with different services initiated at different stages of disease.
- Conventional observational analyses are vulnerable to **time-related bias** and **confounding by disease severity**, making the true survival benefit difficult to quantify.
- Large-scale real-world evidence on which combinations of multidisciplinary services provide the greatest survival benefit remains limited.
- **Target trial emulation** and causal inference offer a framework to compare alternative MDT strategies using longitudinal real-world data.



Data Source & Study Population



GROUSE (GPC Reusable Observable Unified Study Environment)[1] links Medicare fee-for-service claims (2011–2020) for GPC catchment regions with site EHR data harmonized to the **PCORnet CDM**, created in response to PCORI's 2015 call for complete longitudinal ALS, obesity, and breast cancer cohorts.

- **B · Claims:** more complete longitudinal history
- **C · EHR:** richer clinical observations at the point of care
- **A · Crosswalk:** both sources — the complete-case cohort analyzed here

Cohort selection

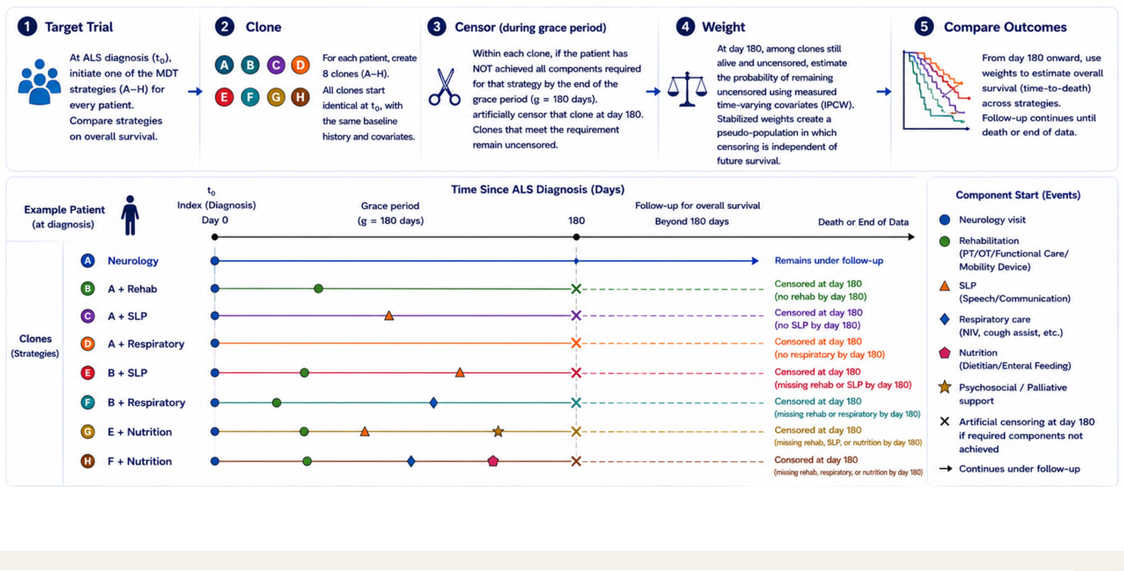
18,472,551 Medicare FFS beneficiaries, 2011 Q1–2020 Q4, ≥1 full year enrolled → **12,490** ≥1 encounter coded for ALS (ICD-9 335.20; ICD-10 G12.21) → **10,777** Confirmed ALS cases (CDC algorithm) → **5,198** With both claims and EHR data (complete cases) → **4,316** Index ALS diagnosis assigned by a neurologist

Case ascertainment (CDC algorithm). Confirmed ALS = ≥1 ALS-coded encounter **plus** ≥1 of: a prescription or dispensing of riluzole, edaravone, or tofersen; a second ALS-coded encounter on a different date; an office visit to a neurologist; or age <65 at Medicare enrollment. **Index date (time zero)** = the earliest ascertainment event.

Study Design — Target Trial Emulation with Clone–Censor–Weight

Target trial emulation (TTE)[2] is a causal inference framework that uses observational real-world data to explicitly mimic the design of a hypothetical randomized trial. **Clone–censor–weight (CCW)**[3] is a TTE implementation for comparing treatment strategies that are established over time: each patient is cloned into every strategy at diagnosis, clones are artificially censored when they deviate from their strategy, and stabilized inverse probability of censoring weights (IPCW) correct for the resulting informative censoring.

Target trial question: does earlier completion of increasingly comprehensive MDT care improve overall survival after the ALS index diagnosis?



MDT strategies (achieved within 180 days)

- **A — Neurology-only (reference):** Neurology visit
- **B — A + Rehab:** Neurology + rehabilitation (PT/OT/functional care/mobility device)
- **C — A + SLP:** Neurology + speech/communication therapy
- **D — A + Respiratory:** Neurology + respiratory care (NIV, cough assist, etc.)
- **E — B + SLP:** Rehab + speech-language pathology
- **F — B + Respiratory:** Rehab + respiratory care
- **G — E + Nutrition:** Rehab + SLP + nutrition/dietitian
- **H — F + Nutrition:** Rehab + respiratory + nutrition/dietitian

How CCW works (left): at t_0 each patient is cloned into strategies A–H. A clone that has not completed all required components by day 180 (grace period) is artificially censored; stabilized IPCW built from measured time-varying covariates create a pseudo-population in which censoring is independent of future survival. Overall survival is compared from day 180 until death or end of data.

Statistical analysis: hazard ratios for death were estimated with an IPCW-weighted marginal structural Cox model, with cluster-robust standard errors at the patient level to account for multiple clones per patient; strategy A is the reference.

Computable Phenotypes of MDT Components

Each component is identified by a **procedure code (CPT/HCPCS)** paired with the **NUCC provider taxonomy** of the provider associated with that code.

Component	Procedure codes (CPT/HCPCS)	Provider taxonomy (NUCC)
Neurology	Office/outpatient E/M visits (99202–99215)	Neurology (e.g., 2084N0400X)
Rehabilitation	PT/OT evaluation & treatment (97161–97168, 97110, 97530); mobility DME	Physical/occupational therapist; PM&R
Speech-language	Speech/swallowing evaluation & treatment (e.g., 92507, 92521–92524)	Speech-language pathologist (e.g., 235Z00000X)
Respiratory	Non-invasive ventilation, cough assist (e.g., E0470, E0471, E0482)	Pulmonology; respiratory therapist
Nutrition	Medical nutrition therapy (e.g., 97802–97804)	Registered dietitian (e.g., 133V00000X)

Covariates

- **Demographics:** age at index, sex, race, ethnicity
- **Socioeconomic:** Medicare–Medicaid dual eligibility or eligibility to low income subsidy
- **ALS phenotype:** diagnoses suggesting distinct ALS phenotypes (e.g., progressive bulbar palsy, [other phenotypes])
- **Disease-modifying therapy:** riluzole, edaravone, tofersen
- **Symptoms:** muscle, speech, walking, swallowing, and movement

Observed Adherence to Assigned MDT Strategy

Strategy	N assessed	N adherent	Adherence
A · Neurology-only	4,316	4,316	100.0%
B · + Rehab	4,316	1,889	43.8%
C · + SLP	4,316	1,327	30.7%
D · + Respiratory	4,316	1,143	26.5%
E · + Rehab + SLP	4,316	1,050	24.3%
F · + Rehab + Respiratory	4,316	624	14.5%
G · + Rehab + SLP + Nutrition	4,316	231	5.4%
H · + Rehab + Resp. + Nutrition	4,316	199	4.6%

All 4,316 eligible patients were assessable for every strategy; 3,561 were observed through the full 180-day grace period. Adherence = N adherent / N assessed.

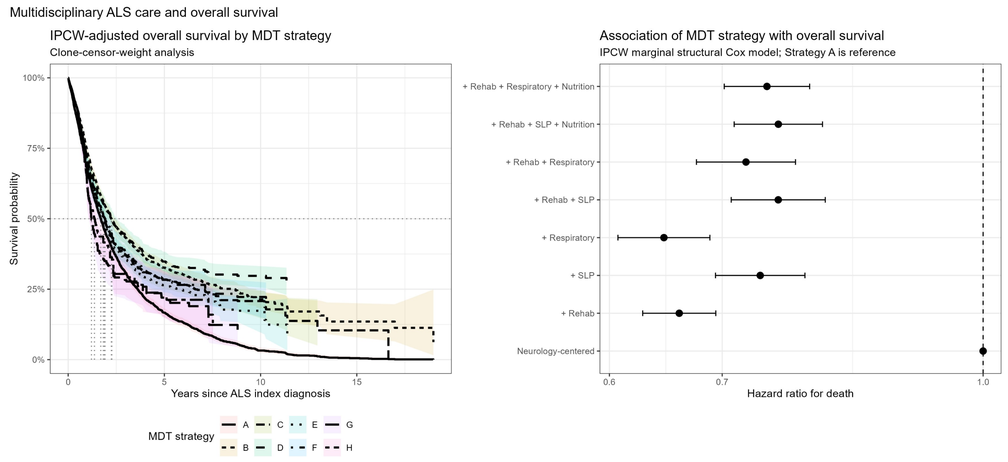
Table 1 — Cohort Characteristics

Characteristic	Overall N = 4,316[1]	Strategy A N = 1,734[1]	Strategies B–H N = 2,582[1]	p[2]
Age at ALS diagnosis, years	69.0 (62.0, 75.0)	68.0 (59.0, 74.0)	70.0 (65.0, 75.0)	<0.001
Age group, years				<0.001
<55	483 (11%)	257 (15%)	226 (8.8%)	
55–64	777 (18%)	372 (21%)	405 (16%)	
65–74	1,918 (44%)	684 (39%)	1,234 (48%)	
75+	1,138 (26%)	421 (24%)	717 (28%)	
Sex				<0.001
Female	2,002 (46%)	717 (41%)	1,285 (50%)	
Male	2,314 (54%)	1,017 (59%)	1,297 (50%)	
Race				<0.001
White	3,742 (87%)	1,456 (84%)	2,286 (89%)	
Black	184 (4.3%)	73 (4.2%)	111 (4.3%)	
Other[3]	58 (1.3%)	26 (1.5%)	32 (1.2%)	
No information	332 (7.7%)	179 (10%)	153 (5.9%)	
Hispanic ethnicity				<0.001
Hispanic	114 (2.6%)	52 (3.0%)	62 (2.4%)	
Non-Hispanic	3,794 (88%)	1,457 (84%)	2,337 (91%)	
No information	408 (9.5%)	225 (13%)	183 (7.1%)	

Characteristic	Overall N = 4,316[1]	Strategy A N = 1,734[1]	Strategies B–H N = 2,582[1]	p[2]
Dual eligibility	339 (7.9%)	153 (8.8%)	186 (7.2%)	0.055
Low-income subsidy	515 (12%)	228 (13%)	287 (11%)	0.040
BMI, kg/m²[4]	26.0 (23.0, 30.0)	26.0 (23.0, 30.0)	27.0 (23.0, 30.0)	0.058
ALS phenotype				
Progressive bulbar palsy	132 (3.1%)	42 (2.4%)	90 (3.5%)	0.053
Primary lateral sclerosis	145 (3.4%)	67 (3.9%)	78 (3.0%)	0.15
Riluzole	406 (9.4%)	154 (8.9%)	252 (9.8%)	0.3
Symptoms				
Muscle movement	1,000 (23%)	319 (18%)	681 (26%)	<0.001
Muscle weakness	1,600 (37%)	502 (29%)	1,098 (43%)	<0.001
Joint/limb pain	2,300 (53%)	786 (45%)	1,514 (59%)	<0.001
Speech	1,587 (37%)	484 (28%)	1,103 (43%)	<0.001
Swallowing	1,573 (36%)	502 (29%)	1,071 (41%)	<0.001
Walking	1,817 (42%)	662 (38%)	1,155 (45%)	<0.001
Baseline labs available				
Any CBC	1,568 (36%)	542 (31%)	1,026 (40%)	<0.001
Any metabolic panel	1,975 (46%)	732 (42%)	1,243 (48%)	<0.001
Follow-up				
Years, median (IQR)	1.3 (0.6, 2.7)	1.6 (0.5, 3.4)	1.1 (0.6, 2.3)	<0.001

[1] Median (IQR) or n (%); patients classified by the most complex strategy achieved within 180 days (descriptive; unassessable patients excluded). [2] Wilcoxon rank-sum; Pearson's chi-squared with simulated p-value (2,000 replicates). [3] Asian, AIAN, NHPI, and multiple races combined to avoid cells <11. [4] Missing for 1,794 (42%). Individual laboratory values (predominantly missing) not shown.

Results — Multidisciplinary Care and Overall Survival



Left: IPCW-adjusted overall survival by MDT strategy (clone–censor–weight analysis). Right: hazard ratios for death with 95% CIs from an IPCW marginal structural Cox model; strategy A (neurology-centered) is the reference. CCW dataset: 4,316 patients · 34,528 strategy clones · 266,090 person-period rows · 2,411 deaths (55.9%) · 13,877 person-years · median follow-up 1.8 years (IQR 0.7–4.5).

Discussion & Conclusion

- All MDT strategies (B–H) were associated with improved survival, with the largest estimated benefits observed for **neurology + respiratory care** and **neurology + rehabilitation (PT/OT)**.
- Some MDT components could not be reliably evaluated because of sparse capture in structured EHR data, potentially under-representing the full spectrum of multidisciplinary ALS care.
- Residual confounding by disease severity remains possible: key ALS severity measures, including FVC and ALSFRS-R, are incompletely captured in structured EHR data.
- Sensitivity analyses are needed to assess robustness to the grace-period definition, alternative or relaxed MDT composition criteria, and IPCW specifications.

Conclusion: These findings highlight the value of target trial emulation using clone–censor–weight methods to evaluate complex, longitudinal ALS care while mitigating time-related biases. Earlier multidisciplinary care was associated with improved survival compared with neurology care alone, although residual confounding, potential exposure misclassification, and limited precision for less common care strategies warrant cautious interpretation. Future studies incorporating richer clinical measures and sensitivity analyses are needed to strengthen causal inference and inform optimal multidisciplinary ALS care strategies.

Acknowledgments

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Poster and figures were generated with AI assistance (ChatGPT, Claude)

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